Student activity 3: unusual compared with what?

Boxplots, regression residuals and influential observations

Group members: replace with your names

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Your task

Prepare a short, reproducible report on a calibration dataset. Your central question is:

> **Which observations deserve closer inspection, and how does that judgement change when we consider an explanatory variable?**

Work in groups of 2–4. Before computing each new analysis, discuss what you expect to learn. Then compare your prediction with the evidence. Keep a short record of conclusions that change as your analysis progresses.

Complete the code chunks and replace the response prompts with your own explanations. The import chunk is ready to run; the analysis chunks contain comments only, so the initial document can be knitted before you complete it. Use either base R graphics or ggplot2, and keep your plotting conventions consistent. Base R is sufficient for every required analysis.

Suggested working time: 90 minutes, followed by a short group discussion.

Study context and data

Imagine a laboratory evaluating a sensor. In each calibration run, a reference input is set to a known level, `x`, and the sensor produces a reading, `y`. The aim is to understand how the readings vary with the input and to identify runs that require further investigation.

This is a **simulated teaching dataset**, framed as a fictional calibration study. It is not a collection of real laboratory measurements or a random sample intended to represent a wider population. The construction details will be discussed after your initial analysis.

Each row represents one run. Row order does not represent time. Three runs, labelled A, B and C, will be followed throughout the activity; these labels do not indicate that a measurement is wrong.

Variable	Meaning	Type / units
`id`	Unique identifier for the calibration run	Character identifier
`x`	Reference input level	Numeric, arbitrary input units
`y`	Sensor reading	Numeric, arbitrary response units

Import the data

Save this Rmd in your working folder and place `boxplots_data.csv` in its `data` subfolder. Keep that relative path when you knit your report. The CSV uses commas to separate fields and a decimal point for numeric values.

```
data_path <- file.path("data", "boxplots_data.csv")
if (!file.exists(data_path)) {
  stop("Place boxplots_data.csv in the data folder next to this Rmd.")
}

dat_raw <- read.csv(data_path, stringsAsFactors = FALSE)
dat <- dat_raw
case_ids <- c("A", "B", "C")

head(dat)
```

```
#>    id         x         y
#> 1 R01 0.0000000  8.804101
#> 2 R02 0.1265823 11.245936
#> 3 R03 0.2531646 10.018843
#> 4 R04 0.3797468 13.574510
#> 5 R05 0.5063291 12.797487
#> 6 R06 0.6329114 13.083398
```

```r
str(dat)
```

```
#> 'data.frame':    83 obs. of  3 variables:
#>  $ id: chr  "R01" "R02" "R03" "R04" ...
#>  $ x : num  0 0.127 0.253 0.38 0.506 ...
#>  $ y : num  8.8 11.2 10 13.6 12.8 ...
```

Keep `dat_raw` unchanged. Add derived variables to `dat` and create a separate copy for the later hypothetical modification. Select observations using `id`, not their row numbers.

If needed, install `knitr` and `rmarkdown` once from the R console. RStudio's **Knit** button renders the document; Pandoc is bundled with RStudio. If you choose ggplot2, install it and load it in your setup chunk.

Step 1: inspect the data and plan your analysis

1. Determine the number of observations and check the variable types.
2. Check for missing or non-finite numeric values and duplicated identifiers.
3. Explain what one row represents and whether there are repeated measurements identified in the supplied data.
4. Agree on an initial plan: which summaries and plots will you use, and what question will each answer?

Useful functions include `nrow()`, `names()`, `summary()`, `is.na()`, `is.finite()` and `anyDuplicated()`.

```r
# Add your data checks here.
```

Your response: report any data-quality issues and list your planned analyses. At this stage, do not discard observations because they look unusual.

Step 2: examine the response on its own

For this step, base your interpretation on `y` alone. Save your conclusions before plotting the relationship with `x`.

1. Summarise the centre, spread and range of `y`.
2. Draw a boxplot of `y` and overlay all observations. Label A, B and C so that they remain distinguishable. Jitter only the categorical plotting position.
3. Obtain the lower and upper box edges, calculate the 1.5-IQR fences, and list the identifiers of observations strictly outside them.
4. Save the resulting logical indicator as `dat$marginal_flag`.

For base R, use the box edges returned by `boxplot.stats()` to match the convention used by `boxplot()`. Its `stats` vector contains the lower whisker, lower box edge, median, upper box edge and upper whisker, in that order. If you use ggplot2, state which quartile convention you use for classification.

The fences are

$$L = Q_1 - 1.5(Q_3 - Q_1), \qquad U = Q_3 + 1.5(Q_3 - Q_1).$$

```
# Calculate descriptive summaries and fences.
# Create dat$marginal_flag and list flagged IDs.
# Draw the boxplot with observations and labels.
```

Your response: which of A, B and C are flagged? Explain what this rule establishes and what further information would be needed before calling a reading a measurement error. Distinguish a fence from a whisker endpoint.

> **Record your initial judgement.** Write one sentence about each of A, B and C using only the evidence from this step. Keep these sentences for comparison with your later conclusions.

Step 3: add the explanatory variable

1. Plot `y` against `x` and label A, B and C.
2. Fit the ordinary least-squares model `y ~ x` using **all observations**. Name the fitted model `fit` and report its intercept and slope.
3. Add the fitted line to the scatterplot. Save its fitted values and residuals as `dat$fitted` and `dat$residual`.
4. For A, B and C, add vertical segments from the fitted line to the observed responses. Explain what their lengths and directions represent.

Useful functions include `lm()`, `coef()`, `fitted()`, `resid()`, `abline()`, `segments()` and `text()`.

```
# Fit the model to every observation.
# Add fitted values and residuals to dat.
# Plot the data, fitted line and labelled residual segments.
```

Your response: revisit your three initial sentences. Which judgements change after considering `x`? Identify any observation whose predictor value extends beyond the range covered by most other runs. What assumption is needed to interpret the fitted line there?

The generating relationship is not supplied at this stage. Base your report on the observed data and fitted model, and distinguish an estimated relationship from a known scientific law.

Step 4: compare response flags with residual flags

For observation i, the residual is

$$e_i = y_i - \widehat{y}_i, \qquad \widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i.$$

1. Draw a boxplot of the residuals with all observations overlaid.
2. Apply the same boxplot convention to the residuals and save the indicator as `dat$residual_flag`.
3. Plot residuals against `x`. Add a horizontal reference at zero and label A, B and C. You may also draw the residual boxplot fences.
4. Make a table for A, B and C containing `id`, `x`, `y`, fitted value, residual, marginal flag and residual flag.
5. Check whether any other observations are flagged and report their IDs.

```
# Calculate residual fences and dat$residual_flag.
# Create both residual plots and the comparison table.
```

Your response: explain the difference between being unusual among all responses and being unusual relative to the fitted relationship. Does a residual boxplot reveal everything visible in the residual-versus-predictor plot? Discuss any pattern in the centre or spread of the residuals.

Residuals are calculated after fitting the model and are not independent copies of its errors. Under the usual constant-variance linear model, their variances depend on leverage. Treat this flagging rule as an exploratory diagnostic, not a formal test with a universal significance level.

Step 5: distinguish residual size, leverage and influence

These quantities answer different questions:

Quantity	Question	Useful R functions
Residual size	How far is the response from the fitted line?	`resid()`, `rstandard()`, `rstudent()`
Leverage	How unusual is the predictor value relative to the others?	`hatvalues()`
Influence	How much does the fitted relationship depend on this observation?	`cooks.distance()`, refitting after omission

1. Calculate leverage, standardized residuals, externally studentized residuals and Cook's distance for all observations.
2. Present a compact table for A, B and C. Also identify the observations with the largest leverage, absolute residual and Cook's distance.
3. Explain whether these rankings agree and what any disagreement tells you.

With an intercept and one predictor,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

Cook's distance combines residual size and leverage. Standardized residuals adjust for leverage and use the overall estimated error scale. Externally studentized residuals estimate that scale with the observation omitted.

```
# Calculate the diagnostic quantities.
# Build a labelled table and compare the rankings.
```

Your response: can a high-leverage observation have a small residual? Must the largest response be the most influential observation? Support your answers with your results. Explain why the diagnostic table does not by itself justify deleting observations.

Step 6: observe influence through case deletion

1. Keep the full-data fit as the reference.
2. Fit three further models, omitting A, B and C **one at a time**. Each fit must retain the other two cases.
3. For each fit, report the intercept, slope and their changes from the full-data coefficients. Define changes as omitted-case fit minus full fit.
4. Calculate the maximum absolute difference in predictions between each reduced fit and the full fit, evaluated at **all original predictor values**.

Use `predict(..., newdata = dat)` to evaluate each reduced model at the same locations. Do not compare fitted-value vectors of different lengths.

```
# Refit after omitting each labelled case separately.
# Compare coefficients and predictions at the original x values.
```

Your response: which omissions change the slope or predictions most? Compare this evidence with Cook's distance. Does any comparison require a reduced model to predict outside its remaining predictor range? This is a sensitivity analysis; it is not a new cleaned dataset.

Step 7: change one response while keeping its predictor fixed

Create a copy named `dat_shift`. Increase A's response by **25 units**, keeping its predictor and every other observation unchanged. Keep the original `dat`.

Before computing: predict whether A's leverage, its residual, its Cook's distance and the fitted slope will change. Give a reason for each prediction.

1. Fit the modified dataset and name the model `fit_shift`.
2. Compare the four quantities between the original and modified scenarios.
3. Fit the data without A and compare the resulting slope across scenarios.
4. Draw two scatterplots using the same axis limits. In each, show the fit including A and the fit excluding A. Make both scenarios fully visible.

```
# Copy the data, then modify A's y value only.
# Refit: previously stored fitted values and residuals are now outdated.
# Recalculate diagnostics from the modified model.
# Create the comparison table and the two scatterplots.
```

Your response: compare the results with your predictions. Explain how an observation can affect the line against which its own residual is measured. Why can ranking observations only by raw residual size be insufficient?

Report and group discussion

Submit the completed Rmd and its knitted HTML. Your report should contain:

- a short description of the data and any issues found during inspection;
- a response boxplot, the fitted relationship, and the two residual displays;

- labelled tables comparing A, B and C, regression diagnostics, case deletion and the modified-A scenario;
- the two plots comparing fits with and without A;
- a conclusion of approximately 150–200 words explaining how context changed your interpretation and what you would check before excluding a measurement.

Figures may be combined into panels. Every figure needs informative axis labels and a caption explaining its analytical purpose. Every important claim should point to a plot, a table or a calculation.

During the group discussion, compare:

1. one initial judgement that changed after introducing `x`;
2. the evidence you used to distinguish residual size, leverage and influence;
3. any differences between groups' plotting or quartile conventions;
4. one limitation of the fitted linear model and one further check you would request from the laboratory.

If you use AI assistance, keep the prompt and describe how you checked the returned code and interpretations. You remain responsible for the analysis.

Optional extensions

- Repeat Step 7 by decreasing A's original response by 25 units. Compare the changes with the upward modification.
- Simulate a curved relationship, fit a straight line and inspect its residuals. Discuss whether apparent anomalies can arise from an inadequate model for the mean.
- Revisit the distribution and sample-size simulations in Unit 05. Explain which cautions about boxplot flags also apply when inspecting residuals.

References

- R documentation: Box Plot Statistics.
- R documentation: Fitting Linear Models.
- R documentation: Regression Deletion Diagnostics.

Note on AI assistance

These teaching materials were prepared with assistance from ChatGPT (OpenAI) in drafting explanations, developing examples and generating R code. The author retains responsibility for the final content.